

The Exact Relative Dimension of the Three-Dimensional Boolean Lattice

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Abstract

We prove that the minimum total element-occurrence cost of a local realizer of the Boolean lattice B_3 is 16, and hence that its relative dimension is 2. The cost-16 upper witness appears in the preprint that introduces relative dimension. For the matching lower bound, we give an explicit nonnegative rational covering dual of value 16. A short argument verifies its inequality for every partial linear extension, so the lower certificate does not depend on an optimizer. An exact-arithmetic exhaustive replay over all 1,323 partial linear extensions provides a second, computational check.

1 Definitions and result

Write $[3] = \{a, b, c\}$ and $B_3 = (2^{[3]}, \subseteq)$. A *partial linear extension* (PLE) M is a linear extension of the subposet induced by some subset of B_3 . A finite family $\mathcal{L}$ of PLEs is a *local realizer* when every comparable pair occurs together in at least one member of $\mathcal{L}$, while every incomparable pair occurs in both relative orders among members of $\mathcal{L}$. The diagonal instances require every element to occur. These are the local-realizer conditions used by Dürschnabel, Hodor, Micek, Stumme, and Trotter [1].

If $V(M)$ denotes the ground set of a PLE, set

$$C(\mathcal{L}) = \sum_{M \in \mathcal{L}} |V(M)|.$$

Since $|B_3| = 8$, the relative frequency of $\mathcal{L}$ is $C(\mathcal{L})/8$, and

$$\text{rdim}(B_3) = \min_{\mathcal{L} \text{ a local realizer}} \frac{C(\mathcal{L})}{8}.$$

Theorem 1. *The minimum total element-occurrence cost of a local realizer of B_3 is 16. Consequently,*

$$\text{rdim}(B_3) = 2.$$

We prove the two inequalities by giving a local realizer of cost 16 and a rational covering certificate with the same value.

2 The upper certificate

Encode a subset of $[3]$ by its three-bit characteristic vector:

$$0 = \emptyset, \quad 1 = a, \quad 2 = b, \quad 3 = ab, \quad 4 = c, \quad 5 = ac, \quad 6 = bc, \quad 7 = abc.$$

Consider the following PLEs, written from first to last:

$$M_1 : 0 \prec 1 \prec 2 \prec 3 \prec 4 \prec 5 \prec 6 \prec 7,$$

$$M_2 : 4 \prec 2 \prec 6 \prec 1 \prec 5 \prec 3,$$

$$M_3 : 5 \prec 2.$$

Each order respects every inclusion relation among its elements. The full linear extension M_1 covers every element and every comparable pair. The singletons occur in the order $1 \prec 2 \prec 4$ in M_1 and in the reverse order $4 \prec 2 \prec 1$ in M_2 . Similarly, the doubletons occur as $3 \prec 5 \prec 6$ in M_1 and $6 \prec 5 \prec 3$ in M_2 . The remaining incomparable pairs are the complementary pairs $1 \parallel 6$, $2 \parallel 5$, and $4 \parallel 3$. Their orders in M_1 are reversed by M_2, M_3, M_2 , respectively. Thus $\{M_1, M_2, M_3\}$ is a local realizer, of cost

$$8 + 6 + 2 = 16.$$

3 The lower certificate

It is convenient to express the local-realizer conditions as covering requirements. Use one requirement for each of the eight elements, one co-occurrence requirement for each of the 19 unordered comparable pairs, and one requirement for each orientation of each of the nine unordered incomparable pairs. There are therefore $8 + 19 + 18 = 45$ requirements. A PLE *covers* the element and comparable-pair requirements that it contains, and it covers the one orientation that it displays for each incomparable pair that it contains.

Lemma 2 (Weighted covering bound). *Suppose nonnegative weights y_r are assigned to the covering requirements and, for every PLE M ,*

$$\sum_{r \text{ covered by } M} y_r \leq |V(M)|.$$

Then every local realizer has cost at least $\sum_r y_r$.

Proof. If $\mathcal{L}$ is a local realizer, every requirement is covered at least once. Nonnegativity and the assumed PLE inequalities give

$$\sum_r y_r \leq \sum_{M \in \mathcal{L}} \sum_{r \text{ covered by } M} y_r \leq \sum_{M \in \mathcal{L}} |V(M)| = C(\mathcal{L}).$$

□

We now specify the weights. Give weight $1/3$ to each of

$$(0, D) \quad \text{and} \quad (D, 7), \quad D \in \{3, 5, 6\},$$

where these are comparable-pair co-occurrence requirements. Give weight $2/3$ to each orientation of every pair of distinct singletons in $\{1, 2, 4\}$ and to each orientation of every pair of distinct doubletons in $\{3, 5, 6\}$. Finally, give weight 2 to the three oriented incomparable requirements

$$6 \prec 1, \quad 5 \prec 2, \quad 3 \prec 4.$$

All other requirements have weight zero. The total weight is

$$6 \binom{1}{3} + 12 \binom{2}{3} + 3(2) = 16.$$

Lemma 3. *For every PLE M of B_3 , the total weight of the requirements covered by M is at most $|V(M)|$.*

Proof. Let s and d be the numbers of singleton and doubleton elements in M , let $e = |V(M) \cap \{0, 7\}|$, and let q count the three weight-2 orientations displayed by M . Directly from the definition of the weights, the load of M is

$$W(M) = \frac{ed}{3} + \frac{2}{3} \binom{s}{2} + \frac{2}{3} \binom{d}{2} + 2q. \quad (1)$$

First, $q \leq 1$. Indeed, for $i \in [3]$ put $D_i = [3] \setminus \{i\}$. If two distinct heavy orientations $D_i \prec i$ and $D_j \prec j$ both occurred, inclusion would force $i \prec D_j$ and $j \prec D_i$. The linear order would then contain the strict cycle

$$D_i \prec i \prec D_j \prec j \prec D_i,$$

which is impossible.

Because $d \leq 3$, the endpoint term satisfies $ed/3 \leq e$. If $q = 0$, then for $u = s, d \in \{0, 1, 2, 3\}$,

$$\frac{2}{3} \binom{u}{2} = \frac{u(u-1)}{3} \leq u.$$

Equation (1) therefore gives $W(M) \leq e + s + d = |V(M)|$.

If $q = 1$, then $s, d \geq 1$. For $u \in \{1, 2, 3\}$ one has $u(4-u) \geq 3$, and hence

$$s(4-s) + d(4-d) \geq 6.$$

Rearranging this inequality yields

$$\frac{2}{3} \binom{s}{2} + \frac{2}{3} \binom{d}{2} + 2 \leq s + d.$$

Together with $ed/3 \leq e$, this again proves $W(M) \leq e + s + d = |V(M)|$. The empty PLE has both load and cost zero. $\square$

Proof of Theorem 1. The three PLEs in the upper certificate give a local realizer of cost 16. The displayed weights have total value 16, and Lemma 3 verifies every PLE inequality required by Lemma 2. Thus every local realizer has cost at least 16. The optimum cost is exactly 16, and division by $|B_3| = 8$ gives $\text{rdim}(B_3) = 2$. $\square$

4 Relation to the inspected literature

Dürschnabel et al. [1] introduce relative dimension and, in Figure 1 and the proof of their Theorem 5, display the same cost-16 local realizer used above to obtain $\text{rdim}(B_3) \leq 2$. Relative to the inspected first version of that preprint, the contribution here is the matching value-16 lower certificate and its universal feasibility proof. This comparison is deliberately limited to that version and is not a claim of global priority.

For a bibliographic example concerning certified bounds in finite-poset counting, see Zeng [2]. No definition, theorem, numerical bound, or originality claim in the present note is inferred from that citation.

5 Exact computational replay

The analytic argument above is the proof of dual feasibility. As a separate reproducibility check, the accompanying Python verifier uses only the standard library and exact rational arithmetic. It generates every linear extension of each of the $2^8 - 1$ nonempty induced subposets by recursively choosing a minimal remaining element. This gives 1,323 PLEs, with counts by size

size	1	2	3	4	5	6	7	8
count	8	37	114	240	348	336	192	48.

It independently filters all 109,600 nonempty partial permutations of the eight elements and obtains the same set. The verifier checks the 1,323 nonempty dual inequalities, the empty inequality, all 45 coverage requirements for the upper witness, and the claimed dual value. From the project directory the deterministic replay command is

```
python3 evidence/verify_certificate.py --output evidence/verification.json
```

Computational and AI assistance. The exact certificate, its checker, and the preparation of this article used assistance from an OpenAI language model. The mathematical conclusion rests on the explicit proof above, and the deterministic checker supplies a replayable secondary verification rather than a solver status or an appeal to AI output. The author reviewed the stated claims and takes responsibility for the manuscript.

References

- [1] Dominik Dürschnabel, Jędrzej Hodor, Piotr Micek, Gerd Stumme, and William T. Trotter, *Relative Dimension of Posets*, arXiv:2609.05166v1 [math.CO], 2026. <https://arxiv.org/abs/2609.05166v1>.
- [2] Zijian Zeng, *Certified Bounds for the Number of Unlabeled Posets on 17 Points*, SSRN working paper no. 7385118. <https://doi.org/10.2139/ssrn.7385118>.